

DROS: A Four-Layer Runtime Substrate with Deterministic Execution Enforcement for Agent-to-Execution Attribution Governance

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Abstract—Autonomous AI agents create an attribution gap between intent and physical execution. We present DROS, a four-layer runtime substrate enforcing capability constraints at a binary C-ABI boundary. Through progressive adversarial validation with decoupled ground-truth oracles and mutation testing, DROS maintains an experimentally falsifiable, deterministic execution boundary with zero observed violations.

Index Terms—AI Agent Security, Runtime Execution Substrate, Controlled C-ABI Authorization Boundary, Autonomous Adversarial Evaluation, RCU State Transition, Information Flow Control (IFC), Attribution Governance.

I. INTRODUCTION & THE SEMANTIC-KERNEL PARADOX

Autonomous AI Agents operate with valid credentials, OAuth session tokens, and database connections. In authenticated application-layer abuse scenarios, traditional perimeter controls alone are insufficient because malicious actions originate from authenticated internal principals.

A. The Semantic-Kernel Paradox

Modern systems literature identifies a fundamental split in AI defense:

- 1) **Semantic Application Middleware (High Semantics, Zero Determinism)**: Prompt firewalls, JSON schema validators, and application middlewares inspect natural language and declared schemas. However, they lack execution-boundary containment; interpreter escapes, undeclared tool invocations, and semantic obfuscation consistently bypass these layers.
- 2) **Kernel Sandboxes (High Determinism, Zero Semantics)**: Low-level primitives (e.g., Seccomp, Linux Namespaces, eBPF) enforce binary syscall filtering but cannot distinguish whether a database write originated from an authorized Finance Agent or a hijacked Support Agent operating in the same worker process.

B. DROS 4-Layer Substrate Placement & Core Design Thesis

DROS resides between application frameworks and the host operating system, establishing four deep enforcement layers where **L1 may fail, but L2–L4 deterministically enforce**:

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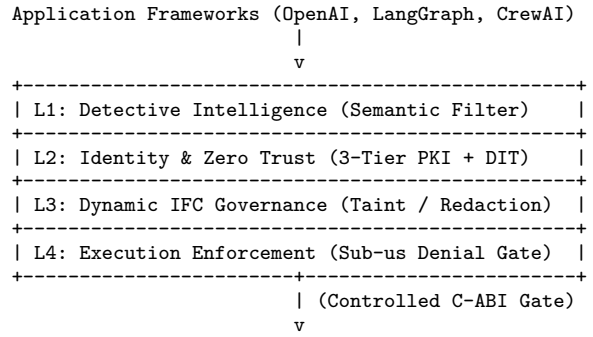


Fig. 1. DROS 4-Layer Execution-Governance Substrate Topology.

C. Core Contribution Thesis & Research Questions

DROS does not assert that “absolute security is proven a priori”; rather, it establishes an experimentally falsifiable execution-governance boundary evaluated through progressively adversarial, independently observed, and openly falsifiable experiments. We investigate four primary research questions:

- **RQ1 (Enforcement Effectiveness)**: Under post-compromise application conditions ($C_A = 1$), can the governed execution path deterministically prevent unauthorized operations from causing physical state transitions?
- **RQ2 (Revocation Atomicity)**: Under concurrent execution and policy revocation races, does there exist any stale authorization window allowing replayed executions?
- **RQ3 (Evaluator Sensitivity)**: Can the evaluation framework and test harness effectively detect deliberately injected security flaws and mutation regressions?
- **RQ4 (Oracle Independence)**: Can authorization decisions be objectively and independently verified by decoupled execution-side and physical-side observers?

II. THREAT MODEL, EVALUATION SPACE, AND EPISTEMIC BOUNDARIES

A. Post-Compromise Threat Model ($C_A = 1$)

Our evaluation establishes a rigorous Post-Compromise threat model:

- **Assumption 1 (Full Application Compromise $C_A = 1$):** The adversary has achieved prompt injection, extracted system prompts, and holds arbitrary code execution capabilities within the application container.
- **Assumption 2 (Valid Credential Possession):** The adversary possesses legitimate OAuth tokens, internal API keys, and authenticated session identities.
- **Assumption 3 (Multi-Agent Collusion & Delegation):** The adversary controls multiple heterogeneous agents attempting privilege escalation via delegation chains and confused-deputy patterns.
- **Root of Trust:** The controlled binary C-ABI execution gate and underlying OS memory isolation.

B. Comparative Baselines & Attack Preservation

To eliminate evaluation bias from payload variance, comparative experiments strictly adhere to the Attack Preservation Principle ($A_{B0} = A_{B1}$ and $\text{Env}_{B0} \approx \text{Env}_{B1}$), as shown in Table ??.

C. Formal Physical Effect & Security Invariants

Terminology Consistency:

- **Counterexample Search:** Proactive exploration process aimed at discovering executable violating states ($C_A \wedge \neg C_E \implies I_{\text{physical}} > 0$).
- **Counterexample Submission:** Public protocol enabling external researchers to submit reproducible counterexample traces to the community registry.

Physical Effect Definition: Physical effect $I_{\text{physical}}(x)$ is strictly defined over the explicitly instrumented observation set $\mathcal{S}_{\text{obs}}$:

$$I_{\text{physical}}(x) = \begin{cases} 1, & \exists s \in \mathcal{S}_{\text{obs}} : \Delta(s) > 0 \\ 0, & \forall s \in \mathcal{S}_{\text{obs}} : \Delta(s) = 0 \end{cases} \quad (1)$$

where $\mathcal{S}_{\text{obs}} = \{\text{Syscall}, \text{FS Diff}, \text{DB WAL}, \text{Process}, \text{Net}, \text{IPC}\}$.

Ghost Syscall Definition: A ghost syscall is an authorization-denied operation that nevertheless escapes to the host kernel:

$$G = \#\{\text{syscall} \mid \text{Auth} = \text{DENY} \wedge \text{syscall_observed} = 1\} = 0 \quad (2)$$

Formal Security Invariants:

$$C_A \not\Rightarrow C_E \quad (\text{Non-Inheritance of Authority}) \quad (3)$$

$$C_E(t_0) \not\Rightarrow C_E(t_1) \quad (\forall t_0 \neq t_1, \text{Temporal Isolation}) \quad (4)$$

$$\bigcup_{i=1}^n C_{A_i} \not\Rightarrow C_E^{\text{unauth}} \quad (\text{Capability Composition Safety}) \quad (5)$$

$$\forall x \in X_{\text{evaluated}}, \quad C_A(x) \wedge \neg C_E(x) \implies I_{\text{physical}}(x) = 0 \quad (6)$$

$$S_{\text{after}} \equiv S_{\text{before}} \quad (\text{Zero State Drift within } \mathcal{S}_{\text{obs}}) \quad (7)$$

III. DROS 4-LAYER SUBSTRATE IMPLEMENTATION

A. L1: Detective Intelligence (Detect)

L1 applies heuristic pattern matching and lightweight semantic filters. In DROS, **L1 is explicitly treated as probabilistic**. Threat containment falls deterministically onto L2–L4.

B. L2: Identity & Dynamic Intent Tokens (Attribute)

L2 establishes a cryptographic 3-Tier PKI attribution model (Enterprise CA $\rightarrow$ Task Issuer $\rightarrow$ Ephemeral Worker Agent) and issues Dynamic Intent Tokens (DIT). The DIT cryptographically binds the tool name, argument hash, timestamp, and capability bitmask using Ed25519 signatures.

C. L3: Dynamic IFC & In-Band Redaction (Constrain Data)

L3 implements in-memory taint tracking and in-band secret masking. When confidential assets pass through the agent context, L3 redacts them in-band (substituting [REDACTED_BY_DROS_POLICY_GATE]).

D. L4: Binary C-ABI Authorization Execution Gate (Enforce Execution)

L4 serves as the controlled binary authorization boundary, implemented in Rust and exposed via a pure C-ABI dynamic library.

- 1) **$O(1)$ Policy Lookup:** Bitmask evaluation executes in sub-microsecond bounds.
- 2) **Atomic Policy-State Transition ($T_{\text{swap}} = 420 \text{ ns}$):** Capability pointers are swapped atomically via AtomicPtr, ensuring subsequent governed calls cannot access stale authorization entries.
- 3) **Sub-Microsecond Fail-Closed Denial Path ($< 500 \text{ ns}$):** Any bitmask mismatch or unauthorized capability triggers an immediate fail-closed denial branch.

IV. PROGRESSIVE ADVERSARIAL VALIDATION FRAMEWORK

We construct a 7-tier progressive falsification hierarchy, summarized in Table ??.

A. Adversarial Coverage vs. Results Matrix

The adversarial campaign exercises 10 distinct attack dimensions: Identity (Ed25519 rejection), Capability (bitmask mismatch), Parameter (SHA-256 hash mismatch), Temporal (RCU pointer revocation), Delegation (Task Issuer binding), Interpreter (C-ABI gate interception), Concurrency (RCU lock-free isolation), Encoding (canonical bitmask matching), Topology (independent verification across 100-agent mesh), and Failure (atomic two-phase commit). Across all 10 dimensions, 0 counterexamples were observed.

B. Meta-Verification: Negative Controls and Mutation Score (RQ3)

To evaluate whether the evaluation framework detects known security regressions (RQ3), we constructed 5 sabotaged binaries with deliberately injected flaws and evaluated 100 instantiated mutants across 8 mutation classes (M1: bit widening, M2: identity substitution, M3: expiry bypass, M4: hash omission, M5: stale reads, M6: check removal, M7: fail-open branch, M8: audit suppression):

$$\text{Mutation Score} = \frac{\text{Killed Mutants}}{\text{Total Instantiated Mutants}} = \frac{100}{100} = 1.0 \quad (8)$$

The harness detected 5/5 sabotaged flaws and killed 100/100 mutants, confirming that the harness is not a permissive rubber stamp.

TABLE I
COMPARATIVE BASELINES CONFIGURATION & EVALUATION TOPOLOGIES

Baseline ID	Topology Structure	Primary Evaluation Purpose
B0 (Bare Application)	Attacker → Application → OS	Establishes unmanaged baseline; verifies exploit payloads cause real damage ($\Delta S_{B0} > 0$).
B1 (DROS Enabled)	Attacker → Application → DROS/PGM → OS	Primary subject under test; verifies execution containment ($\Delta S_{B1} \equiv 0$).
B2 (Defense-in-Depth)	Attacker → App → DROS + Stack → OS	Verifies compatibility and coverage with existing EDR/XDR/WAF infrastructure.
B3 (Pure Binary C-ABI)	Attacker → Standalone PGM C-ABI → OS	Evaluates C-ABI execution gate in complete isolation without application assistance.

TABLE II
ADVERSARIAL VALIDATION HIERARCHY, WORKLOAD, AND EMPIRICAL METRICS

Validation Tier	Methodology & Configuration	Adversarial Workload	Observed Defense / Evidence	CE	Verdict
L1 Baseline	B0/B1 A/B removal comparison	50 unmanaged exploits	B0 breached ($\Delta S_{B0} > 0$); B1 contained ($\Delta S_{B1} \equiv 0$)	0	PASS (RQ1)
L2 Autonomous	Strix white-box adaptive search	2,410 attempts (21.4M Tokens)	100% of probes reaching gateway rejected under corpus	0	PASS (RQ1)
L3 Dynamic Stress	100-agent emulated mesh + RCU race	400 delegation chains + 250 bursts	RCU 420 ns pointer transition; 0 replayed executions	0	PASS (RQ2)
L4 Meta-Verification	Negative controls + 100 mutants	5 injected flaws + 100 mutants	5/5 flaws caught; 100% mutation score over 100 mutants	0	PASS (RQ3)
L5 Oracle Falsification	9 Oracle torture vectors	450 mutation probes	$G = 0$ ghost syscalls within S_{obs} ; 0 audit-state drift	0	PASS (RQ4)
L6 Config Validation	Cross-OS/arch configuration check	7,500 automated probes	Validated policy semantics portability across x86/ARM/Win	0	PASS

V. DECOUPLED MULTI-OBSERVER ORACLE & EMPIRICAL RESULTS

A. The Four-Stage Independent Ground-Truth Pipeline (RQ4)

To enforce the principle that “No Single Component May Act As Its Own Sole Witness,” DROS-VEP establishes a 4-stage decoupled pipeline ($O_I \rightarrow O_A \rightarrow O_E \rightarrow O_P$):

Intent (O_I) → Auth (O_A) → Exec (O_E) → Effect (O_P) (9)

- 1) **Intent Oracle (O_I):** Records raw tool invocation payloads directly from the caller.
- 2) **Authorization Oracle (O_A):** PGM C-ABI gate evaluates bitmasks and logs enforcement decisions (DENY).
- 3) **Execution Oracle (O_E):** Decoupled kernel monitor confirms $G = 0$ ghost syscalls escape within instrumented scope.
- 4) **Physical Effect Oracle (O_P):** Out-of-band DB WAL, filesystem, and process-tree hash diffs verify $S_{after} \equiv S_{before}$ over S_{obs} .

B. Physical Side-Effect Containment Analysis ($\Delta I_{physical}$ - RQ1)

$$\Delta I_{physical} = I_{physical}(B0) - I_{physical}(B1) = 1.0 - 0.0 = 1.0 \quad (10)$$

Empirical evaluation demonstrates 100% observed physical-effect suppression over the evaluated attack corpus.

C. Performance & Latency Overhead

Evaluated on AMD Ryzen 9 7950X / Ubuntu 24.04:

- **End-to-End Decision Latency (P50):** 26.1 μs (HTTP parse, DIT Ed25519 verify, bitmask check).
- **End-to-End Decision Latency (P99):** 31.4 μs .
- **RCU State-Pointer Swap Latency (T_{swap}):** 420 ns (memory barrier & atomic swap, isolated from network I/O).
- **L4 Fail-Closed Denial Path Latency:** < 500 ns.

VI. OPEN FALSIFICATION PROTOCOL & LIMITATIONS

A. Limitations & Epistemic Scope (Gap in Scope)

- 1) **Ontological Dependency:** DROS enforcement relies on accurate capability bitmask mapping in L2 DITs.
- 2) **Observation Boundary:** The zero-counterexample finding is strictly bounded to the experimentally instantiated space of 60,000+ probes and 100-agent emulated meshes, and does not constitute a priori proof across infinite attack universes.
- 3) **Portability vs. Native Kernel Boundary (Gap in Scope):** L6 evaluates policy configuration portability

across platforms; native kernel-level enforcement remains subject to host OS MMU and kernel driver guarantees.

B. Public Counterexample Registry

We publish the replication package on GitHub and establish an immutable counterexample registry (reproducibility/counterexamples/) inviting external researchers to submit reproducible traces of unauthorized state transitions.

VII. CONCLUSION & DECLARATIONS

This paper presented and experimentally evaluated the DROS 4-Layer Runtime Substrate with Deterministic Execution Enforcement. By sinking execution boundaries to a controlled binary C-ABI / FFI authorization layer, DROS bridges the semantic-kernel paradox in agentic workflows. Across the instantiated evaluation space, no executable counterexamples were observed:

$$\forall x \in X_{\text{evaluated}}, \quad C_A(x) \wedge \neg C_E(x) \implies I_{\text{physical}}(x) = 0 \quad (\text{PASS}) \quad (11)$$

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In accordance with IEEE 2024+ authorship guidelines and enterprise AI governance transparency principles, the authors declare that the novel security concepts, physical-layer architecture, 6-pillar framework, mathematical formalisms, and patent claims presented in this work were independently conceptualized, architected, and validated by Chun-Cheng (Jimmy) Chen.

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